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A Parallel Hybrid Quantum-Classical Convolutional Design Using Parameterized Quantum Circuits for Image Classification

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ABSTRACT

Image classification plays a critical role across various industries, yet the rapid growth of visual data has significantly increased computational complexity. Convolutional neural networks (CNNs) have demonstrated strong capabilities in efficient feature extraction, but they still face trade-offs between performance and computational cost when handling large-scale data. Recent studies have explored integrating quantum mechanical principles into classical CNNs to enhance model expressivity. However, existing hybrid quantum–classical convolutional architectures remain limited in terms of feature capture and computational efficiency. In this study, we propose a novel hybrid quantum–classical convolutional neural network (QC-CNN-Parallel), which introduces parallel quantum and classical convolutional branches and extends the concept of multiscale convolution to hybrid networks. This design enables the capture of larger contextual information during convolution while reducing circuit depth and computational overhead. Moreover, we employ parameterized quantum circuits (PQCs) with expressibility, entanglement, and discreteness metrics to select optimal quantum circuits and further enhance model performance. We evaluate QC-CNN-Parallel on three grayscale image datasets: MNIST, Fashion-MNIST, and Overhead-MNIST. Experimental results show that our model improves average training accuracy by 4.89%, 2.77%, and 6.24% over existing hybrid quantum CNNs and classical CNNs, respectively, while exhibiting superior robustness. These results demonstrate that QC-CNN-Parallel achieves enhanced classification performance with high computational efficiency and practical potential.

1 | Introduction

Quantum machine learning (QML) [1] is an interdisciplinary field that merges quantum computing with machine learning. It has garnered significant attention due to its potential to solve computational problems that are challenging or impractical for classical computers. This potential arises from the unique characteristics of quantum computing, such as superposition [2] and entanglement [3], which can provide exponential speedups for specific machine learning tasks.

Researchers have made significant contributions across various domains, including data processing, classification, segmentation, and optimization. In the era of noisy intermediate–scale quantum (NISQ) devices [4], the application of QML to image classification tasks is particularly promising. Quantum computing can expedite the complex computations involved in image analysis, while the classification output remains relatively straightforward and deterministic, making it well suited for the probabilistic outcomes produced by QML algorithms. Several QML models have been proposed for image classification, utilizing techniques such as quantum annealing [5] and PQCs [6, 7].

However, practical implementation of quantum algorithms faces significant challenges, including the fragility of quantum states [8], the necessity of quantum error correction [9], and the effects of noise [10]. Furthermore, it remains an open question whether quantum neural networks (QNNs) [11], a subfield of QML, can consistently outperform classical neural networks. Despite these limitations, QML has demonstrated promising results across several application domains [12].

Hybrid quantum neural networks (HQNNs) [13], as an important variant of QNNs, represent a promising direction for QML in image classification and visual tasks. HQNNs integrate classical deep learning architectures with parameterized quantum circuits (PQCs) to build a hybrid system that leverages both the expressive power of quantum computation and the stability of classical neural networks. They have demonstrated promising applications across multiple domains, including healthcare [14, 15], chemistry [16, 17], data security [18], and finance [19, 20]. Recent studies have further advanced HQNNs in image processing and generative tasks. For instance, He et al. [21] applied HQNNs for image denoising, showing that quantum convolution can enhance texture recovery under high-noise conditions; Huang et al. [22] constructed HQNN-based image classification models and systematically analyzed their robustness against adversarial examples. Moreover, significant progress has been made in quantum generative modeling, including the one-to-many image generation framework based on PQCs proposed by Pei et al. [23], the quantum generative adversarial network (QGAN) leveraging Born machine probabilities by Ding et al. [24], the dual-discriminator QGAN with quantum convolution proposed by Gong et al. [25], and the unrolled hybrid quantum–classical generative model for continuous distribution modeling by Gong et al. [26]. These studies collectively demonstrate the potential of HQNNs in image classification, image generation, and image restoration tasks. Further research is expected to fully explore the representational capacity of HQNNs and to develop more robust and scalable hybrid quantum–classical learning algorithms.

A recent advancement in this field involves the integration of quantum techniques into CNNs. Previous HQNN approaches primarily introduced quantum layers into the fully connected layers to replace traditional linear classifiers. In contrast, the current approach replaces convolutional layers with quantum modules for feature extraction. Recent studies indicate that these models possess significant development potential and already exhibit promising performance. Within this framework, two primary directions have emerged: quantum convolutional neural networks (Q-CNNs) [27], which are entirely quantum architectures, and quantum–classical convolutional neural networks (QC-CNNs) [28], which integrate both classical and quantum components. Each approach has distinct advantages and limitations. QCNNs fully harness the strengths of quantum computing but are less suitable for processing large-scale data. In contrast, QC-CNNs provide improved training efficiency and broader applicability, although they may demonstrate less pronounced enhancements in task-specific performance.

The hybrid QC-CNN was first proposed by Henderson et al. [29] to combine the expressive potential of quantum computing with the well-established capabilities of classical convolutional neural networks (CNNs) for image recognition tasks. Subsequently, Huang et al. [30] introduced a QC-CNN based on variational quantum circuits, validating the effectiveness of this architecture using the MNIST and Fashion-MNIST datasets. Senokosov et al. [31] further extended Henderson's work by replacing the original random quantum circuits with specially designed PQCs, enhancing both the trainability and expressive power of the model. Xiang et al. [32] applied QC-CNNs to breast cancer diagnosis, demonstrating the potential of QML in medical imaging. Later, Shi et al. [33] incorporated quantum convolutional mechanisms into the ResNet framework, proposing the QC-ResNet for medical image classification. Wang et al. [34] systematically evaluated multiple PQC architectures in terms of expressibility and entangling capability, treating these metrics as key quantitative indicators of quantum circuit performance.

Despite the rapid development of QC-CNN-type quantum convolutional models in recent years, several critical challenges remain. Firstly, the quantum convolution proposed by Henderson et al. [29] relies on deterministic or shallow quantum filters without trainable parameters, limiting the ability to exploit quantum nonlinearity for adaptive feature extraction. Although subsequent studies, such as Senokosov et al.'s study [31], introduced trainable PQCs, the quantum modules were often integrated within fully connected layers as independent quantum linear branches, while convolutional feature extraction remained dominated by classical CNNs, restricting their capacity to model multiscale spatial features. Secondly, research such as that by Xiang et al. [32] has primarily focused on applying QC-CNNs to novel tasks rather than systematically optimizing the model architecture. Thirdly, Wang et al. [34] concentrated mainly on single metrics, such as expressibility or entangling capability, overlooking gradient behavior, which is crucial for training feasibility. Moreover, existing methods have not thoroughly addressed additional training challenges associated with increased circuit depth, such as gradient vanishing and the exacerbation of barren plateau phenomena [35]. Collectively, these issues limit the scalability and practical performance of QC-CNNs under NISQ conditions.

To address these challenges, the present study undertakes a systematic investigation and makes three major contributions.

- We propose a parallel hybrid QC-CNN architecture based on PQCs, termed as QC-CNN-Parallel. Inspired by the parallel design concept of Inception networks, this model enhances feature representation by increasing the width of feature extraction layers—implemented as parallel quantum–classical convolutional channels—rather than their depth, effectively mitigating issues associated with excessively deep quantum circuits.
- We systematically investigate the relationship between the structural properties of PQCs and model performance. Our analysis considers multiple dimensions, including expressibility, entangling capability, and gradient behavior, offering both theoretical guidance and practical insights for selecting optimal PQC designs under NISQ conditions.

- The effectiveness and stability of the proposed model are validated through extensive comparative experiments and noise robustness tests. The results demonstrate that QC-CNN-Parallel outperforms the existing quantum and hybrid models in accuracy, training stability, and noise tolerance.

The structure of this paper is organized as follows: Section 2 introduces the relevant background information; Section 3 provides a detailed description of the proposed QC-CNN-Parallel model architecture; Section 4 presents the experimental evaluation of the model's performance; Section 5 summarizes the paper and discusses future research directions; and Table A1 offers a glossary of abbreviations used in this study.

2 | Background

Quantum mechanics defines a "quantum" as the smallest discrete unit of a physical quantity. Core properties of quantum mechanics include superposition and entanglement, which form the theoretical foundation for quantum computing. The quantum bit (qubit) serves as the fundamental unit of information in quantum computing, analogous to the classical bit in conventional computing. Unlike classical bits, qubits can exist in an arbitrary linear superposition of the basis states $|0\rangle$ and $|1\rangle$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1. \tag{1}$$

Quantum entanglement denotes a unique correlation that emerges between two or more quantum systems, wherein the state of one system affects the state of another, irrespective of the distance that separates them. The state of an entangled system cannot be characterized independently; rather, the collective state of the system must be analyzed. For example, in the context of two qubits, the entangled state can be expressed as

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \tag{2}$$

In this specific state, if one qubit is measured and found to be in the state $|0\rangle$, the measurement of the other qubit will also yield the state $|0\rangle$. Conversely, if the first qubit is measured to be in the state $|1\rangle$, the second qubit will similarly be measured as $|1\rangle$.

2.1 | Quantum Logic Gates

In classical computing, the fundamental unit is the bit, and the primary control mechanism is the logical gate, often referred to as the U-gate. By integrating multiple logical gates, one can achieve precise control over electronic circuits. In a parallel manner, quantum computing utilizes the quantum bit, or qubit, as its fundamental operational unit, with the manipulation of qubits being facilitated by quantum logic gates. The implementation of quantum logic gates allows for the deliberate evolution of quantum states, thereby rendering them essential to the development of quantum algorithms.

Quantum logic gates are represented by unitary matrices and generally function on one or two qubits, similar to the function of classical logic gates on bits. This operational mechanism endows quantum computing with considerable flexibility and computational capability. In the following sections, we will introduce several single-qubit quantum gates that will be employed in this study:

$$\begin{aligned}
R_X(\theta) &= \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}, \\
R_Y(\theta) &= \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}, \\
R_Z(\theta) &= \begin{pmatrix} e^{-i(\theta/2)} & 0 \\ 0 & e^{i(\theta/2)} \end{pmatrix}.
\end{aligned} \tag{3}$$

The dual quantum gate represents the dual quantum controlled variant of the previously discussed quantum gate, as outlined below:

$$\mathrm{CU} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{11} & U_{12} \\ 0 & 0 & U_{21} & U_{22} \end{pmatrix}. \tag{4}$$

The operator $\mathrm{U} = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix}$ can represent any single-qubit gate, including, but not limited to, the X, Y, Z, R_Z, R_X, and R_Y gates.

2.2 | Quantum Circuit

Typically, a QNN circuit comprises three primary components: encoding, training, and measurement. The encoding stage is responsible for mapping classical information into quantum states. The training circuit, which incorporates trainable parameters, can be regarded as a quantum analog of a feedforward neural network. It maps input features to corresponding quantum states through parameter optimization. Finally, the measurement stage converts the quantum state back into classical information, which can subsequently be processed by classical networks.

3 | Methodology

Many existing QNN models typically utilize PQCs to substitute for convolutional or classification layers found in conventional neural networks. While increasing the depth of PQCs can improve the model's representational capacity, it often leads to the vanishing gradient problem, which can complicate the training process.

To address this issue, this paper proposes a multichannel hybrid quantum CNN—QC-CNN-Parallel, as illustrated in Figure 1. This architecture integrates classical and quantum convolutional branches in parallel, with the classical branch concentrating on local spatial feature extraction and the quantum branch focusing on nonlinear modeling. This combination enhances the overall representational capacity of the model.

In the concrete implementation, the classical branch employs convolutional kernels of size 44, while the quantum branch

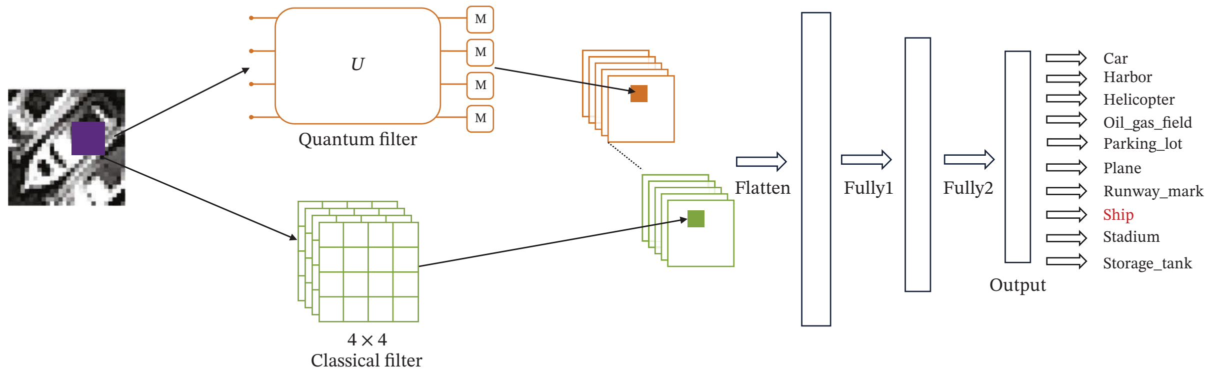


FIGURE 1 | The QC-CNN-Parallel structure proposed in this article.

utilizes kernels of size 22, both with a stride of 2, to achieve efficient spatial compression and feature extraction. To accommodate the receptive field corresponding to the 22 quantum kernel, we design a PQC consisting of four qubits, as illustrated in Figure 2. Specifically, each 22 region within the convolutional window is first flattened into a one-dimensional vector and then encoded into the initial states of four qubits via angle encoding. These qubits are subsequently processed through a trainable PQC composed of multiple layers of parameterized quantum gates, followed by measurement at the output stage. During measurement, the Pauli-Z operator is chosen as observable, and its expectation value is used as the output feature of the quantum convolutional kernel at the corresponding spatial location.

As shown in Figure 2, each qubit corresponds to one output channel. Consequently, the quantum convolutional layer can be viewed as analogous to a classical convolutional kernel with four channels of size 22, where the output of each channel is represented by the measurement result of a single qubit. After the classical and quantum branches independently complete their feature extraction processes, the resulting feature maps are concatenated along the channel dimension and fed into subsequent fully connected layers for classification. This parallel architecture not only alleviates the parameter optimization bottleneck typically encountered in PQC training but also enhances the model's ability to capture multiscale features, thereby demonstrating superior classification performance across multiple tasks. Then, we will introduce the quantum components including feature mapping, PQC, measurement operations, and the classical components and the learning process.

3.1 | Feature Mapping

As image data are inherently classical, it is essential to encode them into quantum states before quantum processing can be applied. Three commonly used encoding strategies in QML are basis encoding [36], amplitude encoding [37], and angle encoding [38]. Each of these strategies has its own trade-offs regarding space complexity, circuit depth, and applicability to specific tasks. These strategies depend on unitary transformations that map classical data vectors into quantum states.

Formally, a classical data vector x is encoded into a quantum state via a unitary operation U_e, such that

$$|\mathrm{x}\rangle = U_e|0\rangle^{\otimes n}, \tag{5}$$

where U_e denotes the encoding circuit composed of quantum gates and n is the number of qubits used.

In this work, we employ angle encoding, which integrates classical data into the rotation angles of quantum gates. As illustrated in Figure 3, each input value is encoded onto a single qubit through a rotation around the Y-axis. Specifically, for an input value x, the corresponding quantum state is prepared as follows:

$$|\psi\rangle = R_Y(x)H|0\rangle = \frac{\cos(x/2)+\sin(x/2)}{\sqrt{2}}|0\rangle + \frac{\cos(x/2)-\sin(x/2)}{\sqrt{2}}|1\rangle. \tag{6}$$

This encoding scheme necessitates one qubit for each input dimension. Although it is less space-efficient than amplitude

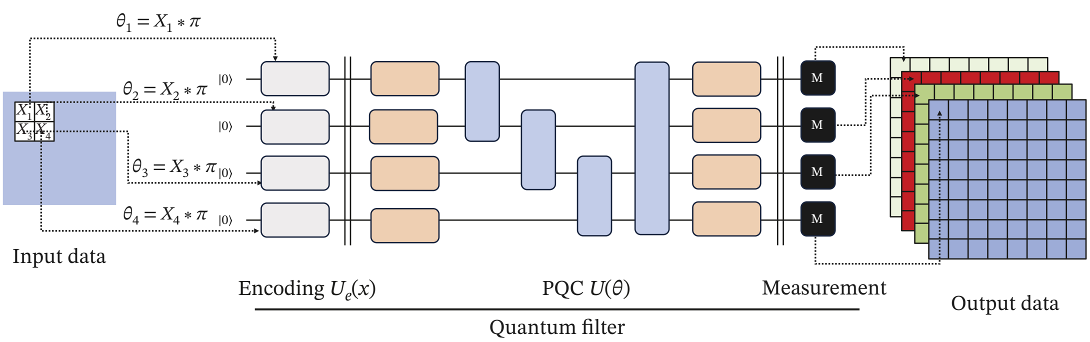


FIGURE 2 | Quantum circuit of feature extraction.

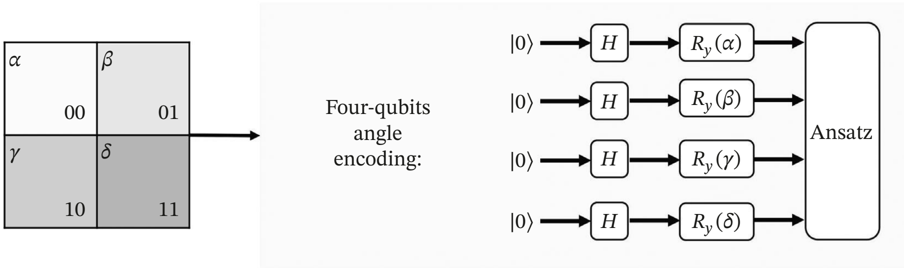


FIGURE 3 | Quantum angle encoding.

encoding, it is more practical and better suited for classification tasks. In our implementation, each input patch of the image is mapped to a set of qubits using this method before being processed by the PQC.

3.2 | PQCs

PQCs represent a critical element of QNNs, and their architectural design can profoundly influence the performance of QNN models. In contrast to encoding layers, which are characterized by fixed functions and lack learning capabilities, the design and selection of trainable layers are essential for effective model construction. These trainable layers play a pivotal role in determining the model's capacity for generalization, similar to the function of convolutional or fully connected layers in classical neural networks. A model that is deficient in tunable parameters may encounter difficulties in learning complex input patterns.

PQCs optimize adjustable parameters to approximate target functions, thereby mapping input features to value ranges that correspond to different classes. The architecture of a PQC is typically composed of two primary components: variational layers and entangling layers. The variational layers generally include single-qubit rotation gates (e.g., R_X, R_Y, and R_Z), which govern the state transformations of individual qubits. In contrast, the entangling layers consist of multi-qubit controlled gates (e.g., CNOT and CZ), which are responsible for establishing entanglement among qubits.

The application of a PQC denoted as $U(\theta)$ to an input state $|\psi\rangle$ results in an output state $|\psi_{\text{out}}\rangle$:

$$J\psi_{\text{out}}\rangle = U(\theta)|\psi\rangle, \tag{7}$$

where $U(\theta)$ can be written as

$$U(\theta) = U_n(\theta_n), \ldots, U_i(\theta_i), \ldots, U_1(\theta_1), \tag{8}$$

where n represents the depth of the circuit, with each parameterized unit that operates on a single qubit being considered as one layer of depth.

Different circuit architectures significantly impact the performance of QNNs, making the selection of an appropriate PQC structure a critical step in developing effective QNNs. Notably, even for the same task, various PQC designs may demonstrate considerable performance variations due to differences in their expressibility and entangling capabilities. As highlighted by Sukin Sim et al. [40], the performance of a PQC is jointly influenced by these two factors. The expressibility of a PQC is typically measured by

$$\text{Expr} = D_{\text{KL}}\left(\widehat{P}_{\text{PQC}}(F;\theta) \| P_{\text{Haar}}(F)\right), \tag{9}$$

where $\widehat{P}_{\text{PQC}}(F;\theta)$ denotes the estimated probability distribution of the fidelity function F, which is derived from states sampled from the PQC with parameter θ. In contrast, $P_{\text{Haar}}(F)$ represents the reference distribution under the Haar measure. The Kullback–Leibler (KL) divergence D_{KL} is utilized to quantify the difference between these two probability distributions. A value closer to zero indicates a higher expressibility of the quantum circuit.

The Meyer–Wallach (MW) entanglement measure [40] is widely utilized to quantify the amount and type of entanglement generated by a PQC, serving as an effective metric for assessing its entangling capability. Specifically, the entangling capability of a PQC is defined as its ability to generate states with high average MW entanglement. This capability is estimated by sampling the parameter space of the PQC and computing the sample mean of the MW entanglement of the output states. The mathematical formulation is provided as follows:

$$\text{Ent} = \frac{1}{|S|}\sum_{\theta_i \in S} Q\left(|\psi_{\theta_i}\rangle\right), \tag{10}$$

where S denotes the set of parameters sampled from the parameter space, $|\psi_{\theta_i}\rangle$ represents the quantum state generated by the PQC with parameter θ_i, and Q is the MW entanglement measure function. Consequently, this metric approaches 1 for circuits with strong entangling capabilities and tends toward 0 for those with weaker entanglement.

However, our empirical observations indicate that neither expressibility nor entangling capability exhibits a positive correlation with the classification performance of the model. Specifically, certain circuits with high expressibility perform unexpectedly poorly, whereas some structures with comparatively low entangling capability achieve higher classification accuracy. This deviation from conventional intuition suggests that relying solely on expressibility and entangling capability is insufficient for comprehensively evaluating PQC quality.

According to the theoretical framework established by McClean et al. [35], the trainability of a PQC is governed by the variance of its gradients. Their central conclusion can be expressed as

$$\text{Var}\left[\partial_\theta L\right] \sim \mathcal{O}!\left(\frac{1}{2^n}\right). \tag{11}$$

Implying that gradient variance decreases exponentially with the number of qubits or circuit depth—an effect commonly referred to as the barren plateau phenomenon. Consequently, the structure of the gradient distribution plays a decisive role in determining whether a PQC is trainable.

Nevertheless, for the shallow and few-qubit quantum convolutional circuits considered in this study, McClean's framework cannot adequately distinguish the training behavior among different shallow PQC architectures. For instance, two circuits may share similar expressibility and comparable mean gradient magnitudes, yet exhibit drastically different training outcomes. This discrepancy suggests the need for an additional metric that captures the variability of gradients across the parameter space—namely, gradient heterogeneity.

To address this, we introduce a new metric termed Discreteness (Disc), defined as the mean variance of the gradients of all trainable parameters over N-random initializations:

$$\mathrm{Var}(g_i) = \frac{1}{N}\sum_{j=1}^{N}\left(g_i^{(j)} - \overline{g}_i\right)^2, \overline{g}_i = \frac{1}{N}\sum_{j=1}^{N} g_i^{(j)}, \tag{12}$$

where $g_i^{(j)}$ denotes the gradient of the i-th parameter during the j-th sampling and N represents the total number of samples. Additionally, $\overline{g}_i$ signifies the mean gradient of the i-th parameter.

$$\mathrm{Disc} = \frac{1}{M}\sum_{i=1}^{M}\mathrm{Var}(g_i), \tag{13}$$

where M represents the total number of trainable parameters.

Fundamentally, this metric captures the second-order statistics of the gradient distribution and aligns with the gradient-variance framework originally proposed by McClean et al., thereby providing a theoretically grounded measure. Detailed numerical simulations and structural comparisons based on these three characteristics will be presented in the experimental section that follows.

3.3 | Measurement

After quantum feature extraction, measurement operations are necessary to convert quantum data into classical information, which is subsequently input into classical networks. The results of the measurements are expressed as expectation values. For a quantum state $|\psi\rangle$, the measurement operator M may be one of the Pauli operators: X, Y, or Z. The measurement outcome is determined by the following.

$$E(\theta) = \langle\psi_{\mathrm{out}}|M|\psi_{\mathrm{out}}\rangle = \langle\psi|U(\theta)^{\dagger}\mathrm{MU}(\theta)|\psi\rangle. \tag{14}$$

In this experiment, we use the Pauli-Z operator as observable. This choice is motivated by the fact that the eigenstates of Pauli-Z, namely, $|0\rangle$ and $|1\rangle$, correspond to the default computational basis of quantum computers, allowing for direct physical measurement at the hardware level. The measurement outcomes naturally represent the probability distribution of a qubit between the $|0\rangle$ and $|1\rangle$ states. Furthermore, the eigenvalue of Pauli-Z is ± 1, which restricts its expectation values to the interval $[-1, 1]$. This bounded range is stable, easily normalized, and therefore particularly suitable as input features for classical neural networks. Due to these advantages, Z-basis measurement not only offers simplicity and compatibility in implementation but also exhibits robustness in error mitigation and numerical stability, making it widely adopted in QNNs and related quantum computing applications.

3.4 | Classical Dense Layer

After obtaining the feature data from the PQC output, further processing is conducted. In the hybrid layer of the model, features extracted by the quantum convolutional layer (PQC) are concatenated with those from the classical convolutional layer along the channel dimension, resulting in multichannel feature fusion. This fusion strategy effectively harnesses the complementary strengths of both quantum and classical features, thereby enhancing the model's representational capacity.

The fused features are subsequently fed into a three-layer fully connected neural network for the purpose of classification. A ReLU activation function is employed between each pair of fully connected layers to augment the model's ability to capture nonlinear relationships. The final classification output is trained utilizing the cross-entropy loss function, which serves to optimize the model parameters.

3.5 | Learning Process

The efficacy of quantum algorithms that utilize PQCs is significantly influenced by the efficiency and robustness of the optimization techniques applied. Following the execution of measurements and the acquisition of output features from the quantum circuit, these features are subsequently transmitted to classical fully connected layers, where gradient computation and parameter updates are performed.

In this study, we utilize the cross-entropy loss function as the optimization objective, which is defined as follows:

$$\mathcal{L}(\theta) = -\sum_{i=1}^{C} y_i \log\left(\frac{e^{E(\theta_i)}}{\sum_{j=1}^{C} e^{E(\theta_j)}}\right), \tag{15}$$

where $E(\theta_i)$ represents the output of the quantum circuit for class i given the parameter θ (refer to Equation (14)). The variable y_i denotes the one-hot encoded ground truth label, while C signifies the total number of classes involved in the classification task. The gradient of the loss function with respect to the parameter θ_i is given by

$$\frac{\partial\mathcal{L}(\theta)}{\partial\theta_i} = \sum_{i=1}^{C}(\sigma_i - y_i)\frac{\partial E(\theta)}{\partial\theta_i}, \tag{16}$$

where $\sigma_i = e^{E_i}/(\sum_{k=1}^{C} e^{E_k})$ denotes the predicted probability for class i obtained by softmax normalization. To calculate the partial derivative of the quantum circuit output $E(\theta)$ with respect to the parameter θ_i, we utilize the parameter-shift rule, which is articulated as follows:

$$\frac{\partial E(\theta)}{\partial\theta_i} = \frac{1}{2}\left[E\left(\theta + \frac{\pi}{2}e_i\right) - E\left(\theta - \frac{\pi}{2}e_i\right)\right], \tag{17}$$

where e_i denotes the unit vector associated with the i-th parameter. Based on the gradient information presented above, the parameters can be updated utilizing the gradient descent method. The update rule is articulated as follows:

$$\theta_i' = \theta_i - \eta\frac{\partial\mathcal{L}(\theta)}{\partial\theta_i}, \tag{18}$$

where η represents the learning rate, which regulates the magnitude of each parameter update. This methodology is generally implemented using mini batches instead of the complete training

TABLE 1 | Experimental data for performance evaluation.

Data	Categories	Patch size	Training	Test
MNIST	All classes	$28 \times 28 \times 1$	10,000	2000
Fashion-MNIST	All classes	$28 \times 28 \times 1$	10,000	2000
Overhead-MNIST	All classes	$28 \times 28 \times 1$	8519	1065

dataset, thereby mitigating the risk of converging to suboptimal local minima.

4 | Results

4.1 | Experiments' Setting

In this chapter, we design and conduct three experiments, detailed as follows.

- Experiment 1: This experiment aims to investigate the relationship between the structural characteristics of various training circuits and model performance. We seek to identify the most appropriate training circuit for this study and analyze the key factors that influence the model's effectiveness.
- Experiment 2: Building on the optimal training circuit identified in Experiment 1, this experiment incorporates it into our proposed model architecture and compares the resulting model with existing quantum–classical hybrid models. The objective is to assess and validate the performance of various models under consistent conditions.
- Experiment 3: This experiment evaluates the robustness of the models by introducing various types of noise, including both data perturbations and model disturbances. The objective is to analyze each model's resistance to interference and its overall robustness in noisy environments.

4.2 | Experiments' Preparation

4.2.1 | Data Preparation

In this study, three datasets—MNIST, Fashion-MNIST, and Overhead-MNIST—were utilized. To evaluate model performance under typical operational conditions, all categories were retained without any class reduction. Due to the computational limitations of current quantum simulators, we applied class-balanced subsampling to MNIST and Fashion-MNIST by randomly selecting 1000 samples per category for training and 200 samples per category for testing, ensuring controlled dataset size and balanced class distributions. The full Overhead-MNIST dataset was used, as its original scale is already suitable for experimentation. A summary of the data preparation procedures and experimental configurations is provided in Table 1.

4.2.2 | Quantum Simulator Setup

In this study, we implemented and trained quantum deep learning models using the PennyLane platform [39], a widely adopted framework for QML that supports multiple quantum simulators and hardware backends. For gradient-based optimization, we employed PennyLane's PyTorch interface, specifically the TorchLayer module, which encapsulates quantum nodes (QNodes) as custom layers fully compatible with PyTorch. This integration allows trainable parameters within quantum circuits to be optimized using PyTorch's automatic differentiation framework. Moreover, TorchLayer facilitates the construction of hybrid quantum–classical architectures by seamlessly combining quantum circuits with conventional neural network layers, such as convolutional and fully connected layers, enabling joint optimization of classical and quantum parameters.

4.3 | Experiments' Result

4.3.1 | Experiment 1: Analysis of the Performance Indicators of Different Quantum Circuits

In this experiment, the objective is to identify quantum circuits that exhibit enhanced overall performance. To minimize the impact of confounding variables, we have developed a straightforward hybrid model that comprises a single quantum convolutional layer for feature extraction and a classical linear layer for classification. This simplified architecture substantially decreases the number of parameters, thereby reducing the likelihood of overfitting on relatively small datasets and facilitating a more precise assessment of the contribution of the quantum component.

Figure 4 illustrates nine representative quantum circuit structures, each constructed by combining common single-qubit rotation gates (R_X, R_Y, R_Z) with three types of two-qubit entangling connectivities: Linear, Circle, and All-to-All. Additionally, for performance comparison, we include two more complex circuit architectures proposed by Sim et al. [40], which have demonstrated strong empirical effectiveness; these are shown in Figure 5. The corresponding gate-level definitions are provided below:

Circuit-10 consists of three components: the first variational layer, the entangling layer, and the second variational layer.

- Variational layers: Each qubit is parameterized by two rotation gates. For an $n-$qubits system, a rotation layer is defined as

$$U_{\mathrm{rot}}(\theta) = \prod_{i=1}^{n} R_X\left(\theta_i^{(1)}\right) R_Z\left(\theta_i^{(2)}\right). \tag{19}$$

- Entangling layer: Every ordered pair of distinct qubits (i, j) is connected through a CRX gate, where qubit i serves as the control qubit and qubit j serves as the target qubit:

$$U_{\mathrm{ent}}(\phi) = \prod_{i=1}^{n} \prod_{j=1; j \neq i}^{n} \mathrm{CRX}_{i \longrightarrow j}\left(\phi_{ij}\right). \tag{20}$$

Circuit-10 is expressed as

$$U_{10}(\theta, \phi) = U_{\mathrm{rot}}^{(2)}(\theta_2) U_{\mathrm{ent}}(\phi) U_{\mathrm{rot}}^{(1)}(\theta_1). \tag{21}$$

Circuit-11 follows a similar structure but employs a Circle entanglement pattern.

- Variational layers: Each qubit is parameterized by a single rotation gate:

$$U_{\mathrm{rot}}(\theta) = \prod_{i=1}^{n} R_Y(\theta_i). \tag{22}$$

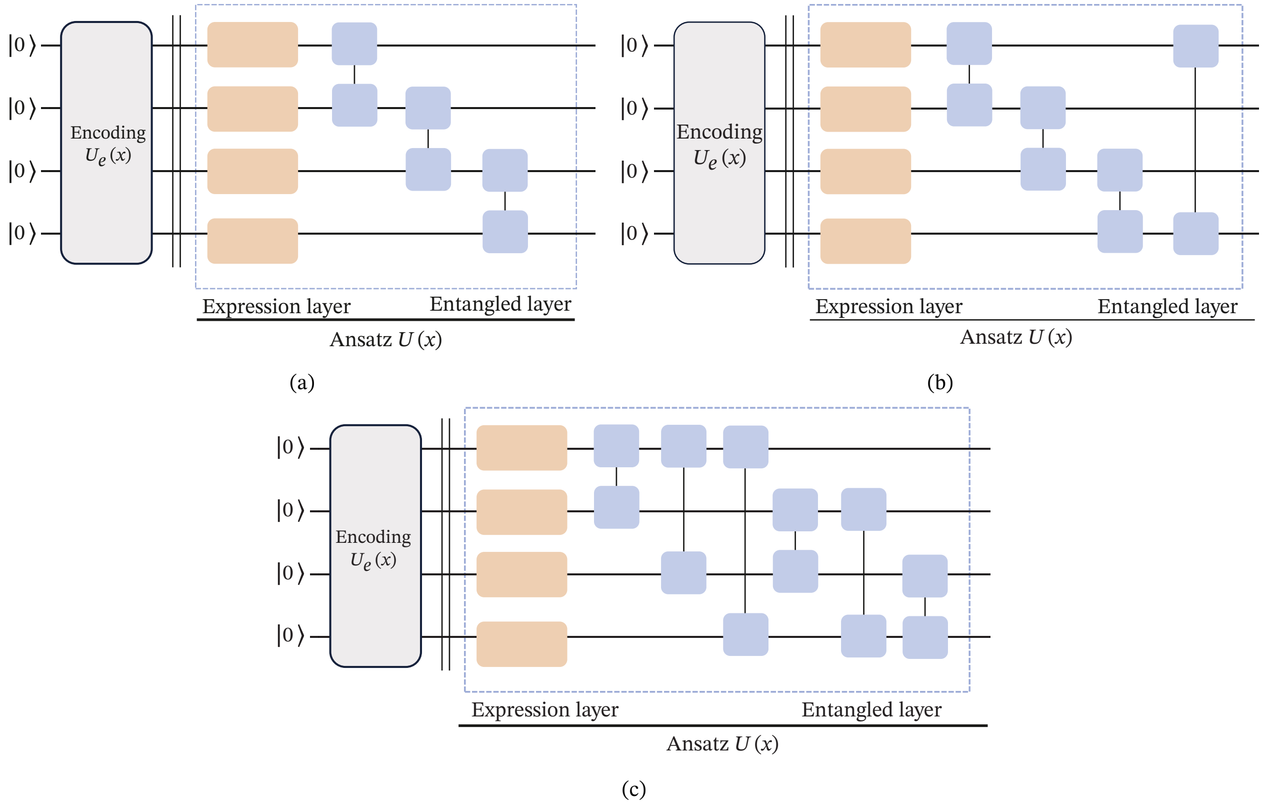


FIGURE 4 | This study compares three classical two-qubit entangling circuit architectures used as training circuits. (a) Linear. (b) Circle. (c) All-to-All.

- Entangling layer: The entanglement is arranged in a ring topology in which each qubit q_i controls a CRX gate acting on its successor qubit $q_{(i \bmod n)+1}$:

$$U_{\text{ent}}(\phi) = \prod_{i=1}^{n} \text{CRX}_{i \longrightarrow (i \bmod n)+1}(\phi_i). \tag{23}$$

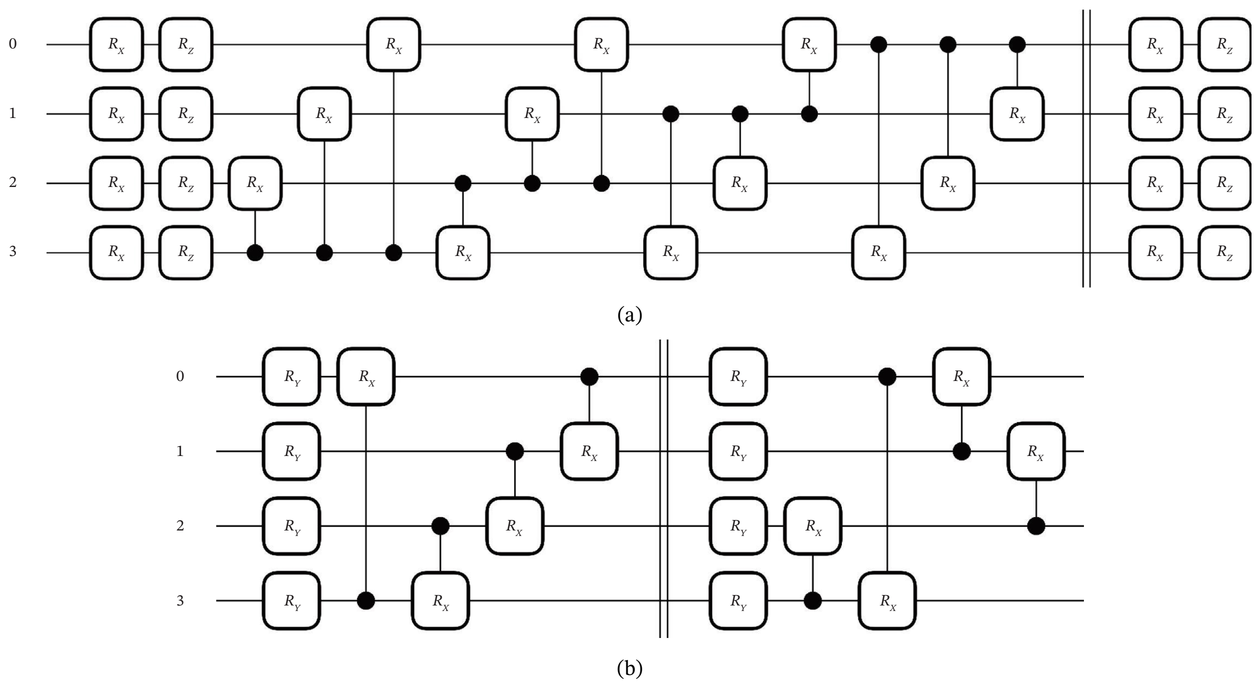


FIGURE 5 | The two additional entangling structures selected for this study. (a) Circuit 10. (b) Circuit 11.

TABLE 2 | Expressibility and entangling capability of different construction.

Circuit	Params	Type	Expr	Ent	Disc
R_X	4	Linear	0.1755	0.5618	0.0280
		Circle	0.1679	0.7448	0.0060
		All-to-All	0.1738	0.6178	0.0330
R_Y	4	Linear	0.3317	0.4540	0.1256
		Circle	0.3552	0.6372	0.0607
		All-to-All	0.3454	0.4520	0.1260
R_Z	4	Linear	0.1721	0.6248	$2.7e^{-33}$
		Circle	0.1670	0.7924	$3.6e^{-33}$
		All-to-All	0.1759	0.5653	0.0154
Circuit 10	28	—	0.0013	0.7180	0.0208
Circuit 11	16	—	0.0071	0.5463	0.0191

Note: The Type indicates the three circuit topologies illustrated in Figure 4. Lower expressibility reflects better uniformity over the Hilbert space.

Circuit-11 is expressed as

$$U_{11}(\theta, \phi) = U_{\text{ent}}^{(2)}(\phi_2)U_{\text{rot}}^{(2)}(\theta_2)U_{\text{ent}}^{(1)}(\phi_1)U_{\text{rot}}^{(1)}(\theta_1). \tag{24}$$

where both entangling layers adopt a Circle topology, but they differ in their starting control qubit: The first layer begins with qubit q_1, while the second starts with qubit q_2. This shift prevents the two entangling patterns from fully overlapping and enhances the expressive capacity of the circuit.

In the context of the experimental setup, each quantum circuit architecture was constructed utilizing four qubits and subjected to 5000 numerical simulations to thoroughly evaluate its expressibility, entangling capability, and circuit diversity. The resultant data are presented in Table 2. As indicated in the table, circuits that incorporate identical types of single-qubit rotation gates demonstrate comparable levels of expressibility; however, the observed variations in entangling capability are predominantly ascribed to differences in the topology of the two-qubit entangling connections.

A thorough comparison of various configurations reveals the following observations: with respect to expressibility, the trend is observed as $R_X \approx R_Z > R_Y$; regarding entangling capability, the hierarchy is Circle > Linear > All − to − All.

Table 2 presents the circuit diversity associated with each configuration. Following this, we integrate the 11 quantum circuits into the previously mentioned hybrid model and perform classification experiments utilizing standard benchmark datasets. The classification accuracies are summarized in Table 3. The experimental results indicate that, although the All-to-All architecture demonstrates relatively limited entangling capability, its high diversity significantly enhances performance. Similarly, circuits that utilize the R_Y gate, despite their restricted expressibility and entangling capability, benefit from increased diversity, which aids the optimization process by providing more informative gradients, thereby resulting in improved classification accuracy.

It is important to highlight that Circuit 10 and Circuit 11 demonstrate an advantageous balance between expressibility and diversity. These circuits integrate robust representational capabilities, significant diversity, and distinct discriminability, which collectively contribute to their exceptional overall performance in the conducted experiments.

In summary, when designing hybrid Q-CNNs, the selection of circuit structure should comprehensively take into account three key factors: expressibility, discreteness, and distinguishability. The experimental results indicate the following:

- Circuits constructed with R_Y gates, despite their limited expressibility, demonstrate high discreteness, resulting in stable training and enhanced classification performance.
- Circuits that utilize R_X and R_Z gates exhibit greater expressibility; however, they experience reduced discreteness, making them susceptible to barren plateaus and consequently leads to diminished performance.
- Among entangling structures, the Circle architecture exhibits a stronger entangling capability but has poorer distinguishability.
- Customized architectures, such as Circuit 10 and Circuit 11, achieve a superior balance among expressibility, discreteness, and distinguishability. Although they require slightly more parameters, they deliver the best performance.

These analyses indicate that the key to constructing high-performance hybrid quantum models lies in the meticulous design of circuit structures, which necessitate a careful balance between model complexity and trainability, tailored to specific tasks.

4.3.2 | Experiment 2: Analysis of General Classification Performance

Table 4 summarizes the configurations of the models compared in this study. In this experiment, the optimal PQCs identified in the first set of experiments were integrated into the hybrid model architectures, and their performance was compared against a classical CNN and several existing hybrid quantum models. Specifically, the classical CNN baseline adopts the conventional LeNet-5 architecture; the QC-CNN model follows the design of Henderson et al. [29]; the HQNN-Quanv model is based on the scheme proposed by Senokosov et al. [31]; the VCNN model comes from Huang et al. [30]; the QC-ResNet model from Shi et al. [33]; and the QC-Inception model from Liu et al. [34]. The primary distinction among these hybrid models lies in the trainability and structure of the quantum circuits: The QC-CNN employs fixed (nontrainable) circuits, HQNN-Quanv utilizes trainable PQCs, while VCNN, QC-ResNet, and QC-Inception incorporate different quantum–classical convolutional designs to enhance feature extraction.

Subsequently, we trained and evaluated these models on multiple datasets, as described in Section 4.2.1. The results are presented in Figures 6, 7, and 8.

From the figures, it can be observed that under consistent experimental conditions, our proposed model achieves the highest classification accuracy across all datasets while using the fewest parameters. Upon analysis, this advantage primarily stems from the following factors.

- Multichannel architecture design: Our model utilizes parallel classical and quantum channels, facilitating

TABLE 3 | Classification performance of different constructions.

Circuit	Type	MNIST-acc	Fashion-MNIST-acc
R_X	Linear	0.6560	0.7364
	Circle	0.6066	0.6864
	All-to-All	0.7495	0.7647
R_Y	Linear	0.7530	0.7626
	Circle	0.7602	0.7763
	All-to-All	0.8006	0.7846
R_Z	Linear	0.5496	0.6229
	Circle	0.5481	0.6213
	All-to-All	0.7249	0.6881
Circuit 10	—	0.8254	0.7946
Circuit 11	—	0.8057	0.7778

complementary feature extraction from various scales and perspectives, which enhances the overall representational capacity.

- Introduction of the optimal PQC: The selected PQC is determined through prior analysis to achieve a well-balanced trade-off among expressibility, discreteness, and entangling capability.
- Reasonable control of model depth: Compared to other models that are deeper and more difficult to train, our design maintains a relatively shallow architecture while still achieving robust representational power.

Overall, the proposed QC-CNN-Parallel architecture leverages the complementary strengths of quantum and classical feature extraction while using relatively few parameters, demonstrating superior classification performance.

4.3.3 | Experiment 3: Analysis of the Effects of Noise on Model Performance

To thoroughly assess the robustness of the proposed model in the presence of noisy conditions, we performed a series of performance tests across different types of noise and compared the outcomes with those of traditional convolutional models. In terms of data noise, to effectively isolate the effects of noise from the dimensionality reduction process, Gaussian noise was introduced to the input images subsequent to dimensionality reduction. This approach more accurately simulates the influence of noise at the perceptual layer on the performance of the model.

In addressing the noise inherent to the model, we examined three distinct types of noise that are applied immediately prior to the measurement of qubits: bit-flip noise, phase-flip noise, and depolarizing noise. The definitions of these noise types are as follows.

- **Bit-Flip Noise**: Bit-flip noise is one of the fundamental quantum error models and corresponds to the application of a Pauli-X gate on a qubit with probability p. This noise causes the quantum state to randomly flip between $|0\rangle$ and $|1\rangle$. Its evolution can be described as

$$\rho \longrightarrow (1-p)\rho + pX\rho X, \tag{25}$$

where X is the Pauli-X operator.

- **Phase-Flip Noise**: Phase-flip noise modifies the phase of the quantum state by applying a Pauli-Z gate with probability p. When acting on a qubit, the phase of the $|1\rangle$ state is inverted while the $|0\rangle$ state remains unchanged. Its mathematical expression is

$$\rho \longrightarrow (1-p)\rho + pZ\rho Z, \tag{26}$$

where Z is the Pauli-Z operator.

- **Depolarizing Noise**: Depolarizing noise is a widely used general error model in quantum channels. With probability p, it replaces the quantum state with a maximally mixed state, and with probability $1-p$, it leaves the state unchanged. Its evolution is given by

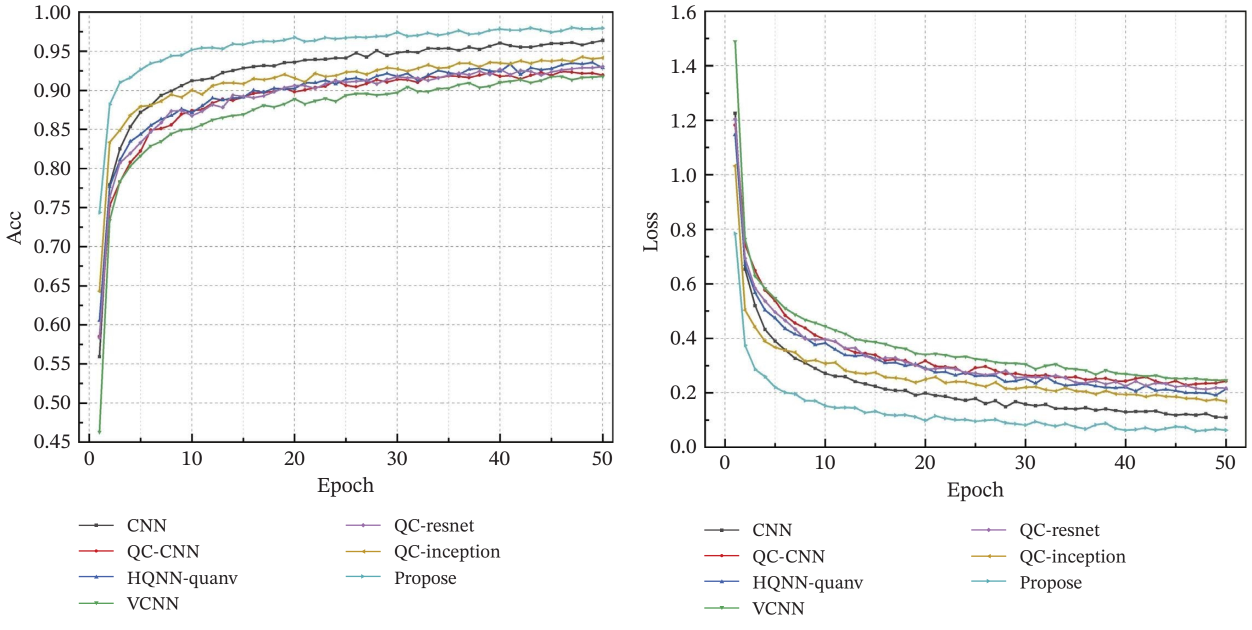
$$\rho \longrightarrow (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z), \tag{27}$$

where X, Y, and Z are the Pauli operators.

TABLE 4 | The experimental configuration of different models.

Indicators	Model						
	CNN	QC-CNN	HQNN-Quanv	VCNN	QC-ResNet	QC-Inception	Proposed
Parameters	464	448	448	456	512	304	136
Qubits	\	4	4	4	4	4	4
Receptive field	4	4	4	4	4	4	4
Depth	4	4	4	4	4	3	3
Learning rate	0.01						
Random seed	42						
Loss function	Cross-entropy loss						
Batch size	32						
Epoch	50						
Optimizer	Adam						

Note: Since all models use the same linear layer, the table only includes the number of parameters in the convolutional part.

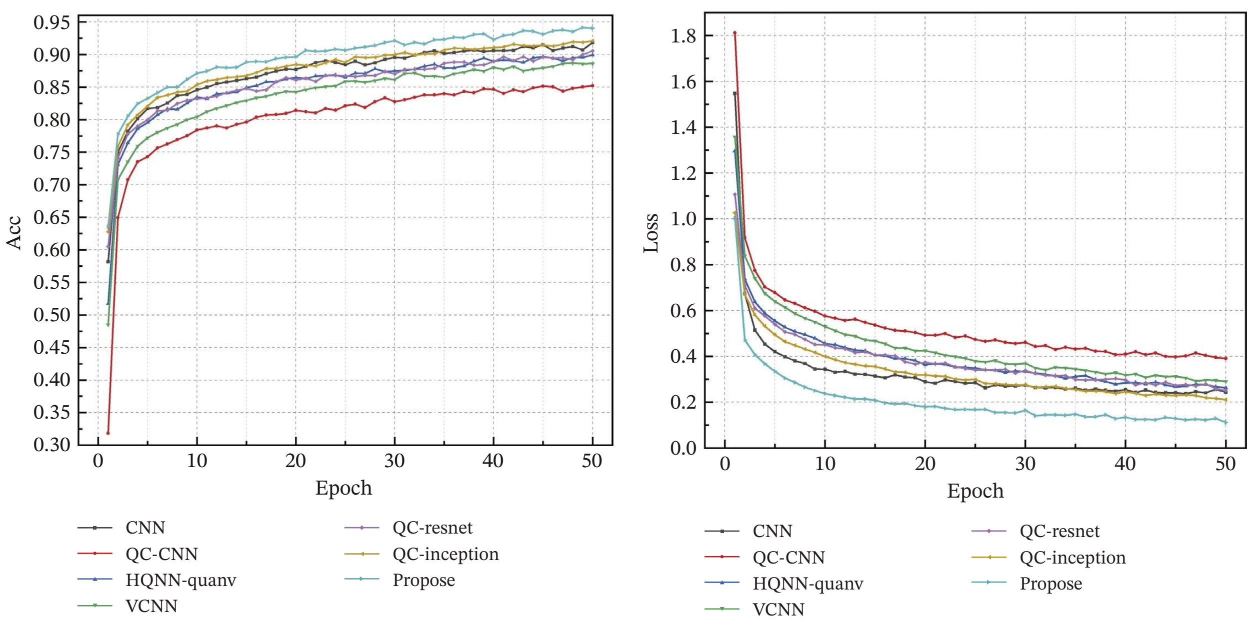


(a) (b)

FIGURE 6 | Validation performance of various models on the MNIST dataset over training iterations: (a) accuracy and (b) loss.

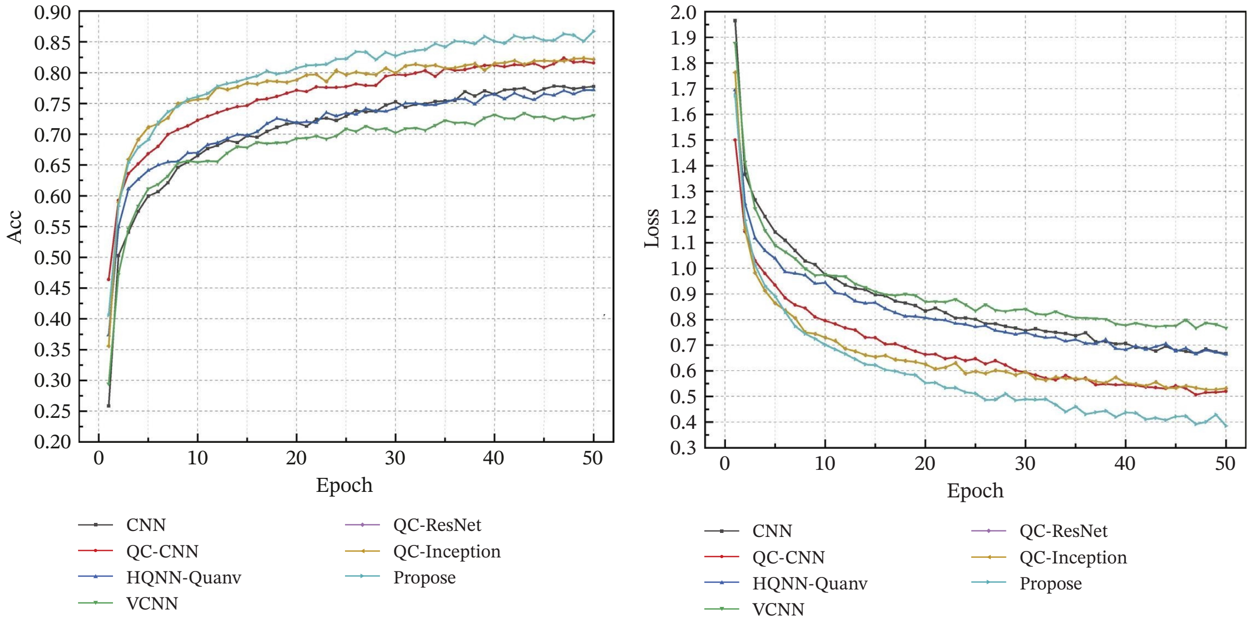
In this experiment, we employed noise characterized by three distinct probability coefficients ($p = 0.1, 0.2, 0.3$) to systematically assess the robustness of the proposed quantum generative model under varying levels of noise intensity. To simulate a noisy quantum environment, we utilized the mixed-state quantum simulator (default.mixed) provided by the PennyLane framework. However, it is important to note that this simulator currently lacks support for batched inputs, which significantly escalates the computational cost associated with model training. Given this limitation, the experimental task was appropriately simplified to concentrate exclusively on the analysis of noise sensitivity within the MNIST dataset, encompassing all categories.

In the process of model selection, we evaluated the previously introduced models: QC-CNN, HQNN-Quanv, the proposed model, and the classical CNN. To ensure a fair evaluation and accurate performance comparison, all models were trained and tested on the complete validation and test sets corresponding to the categories outlined in Table 1. The experiments were conducted over 50 training epochs with a batch size of 100. The



(a) (b)

FIGURE 7 | Validation performance of various models on the fashion-MNIST dataset over training iterations: (a) accuracy and (b) loss.



(a)

(b)

FIGURE 8 | Validation performance of various models on the overhead-MNIST dataset over training iterations: (a) accuracy and (b) loss.

TABLE 5 | Classification performance comparison with the data noise.

Model	No noise	Data noise		
		Error rate 0.1	Error rate 0.2	Error rate 0.3
Propose	0.9005	0.8915	0.8900	0.8425
HQNN-Quanv	0.8320	0.8344	0.7796	0.7125
QNN	0.8350	0.8170	0.7544	0.7100
CNN	0.8935	0.8840	0.8530	0.7844

TABLE 6 | Classification performance comparison with the bit-flip noise.

Model	No noise	Bit-flip noise		
		Error rate 0.1	Error rate 0.2	Error rate 0.3
Propose	0.9005	0.8769	0.8558	0.8405
HQNN-Quanv	0.8320	0.6775	0.6523	0.6399
QNN	0.8350	0.7115	0.6124	0.4615

experimental results are presented in Tables 5, 6, 7, and 8. The primary conclusions drawn from these results are as follows.

- **Noise-Free Condition**: In the absence of noise, the proposed model attains an accuracy of 0.9005, which represents a significant improvement over the HQNN-Quanv, QNN, and classical CNN baseline models.

- **Robustness to Data Noise**: The proposed model demonstrates significant robustness in the presence of classical data noise, with error rates ranging from 0.1 to 0.3. Notably, even at the highest error rate of 0.3, the model achieves an accuracy of 0.8425. In contrast, the accuracies of the HQNN-Quanv and QNN models decline to 0.7125 and 0.7100, respectively.

- **Impact of Bit-Flip Noise**: The presence of bit-flip noise in quantum systems has a substantial impact on the performance of HQNN-Quanv and QNN models, particularly in high-noise environments characterized by an error rate of 0.3. Under these conditions, the accuracies of the HQNN-Quanv and QNN models decline to 0.3340 and 0.4615, respectively. In contrast, the proposed model exhibits a notable accuracy of 0.8405 under identical conditions, thereby demonstrating a robust resilience to qubit bit-flip errors.

- **Impact of Phase-Flip Noise**: All models demonstrate a relatively stable robustness in the presence of phase-flip noise. The proposed model attains an accuracy exceeding 0.8600 at an error rate of 0.3, suggesting a notable degree of tolerance to phase decoherence.

TABLE 7 | Classification performance comparison with the phase-flip noise.

Model	No noise	Phase-flip noise		
		Error rate 0.1	Error rate 0.2	Error rate 0.3
Propose	0.9005	0.8836	0.8618	0.8602
HQNN-Quanv	0.8320	0.8279	0.8254	0.8267
QNN	0.8350	0.8272	0.8191	0.8167

TABLE 8 | Classification performance comparison with the depolarizing noise.

Model	No noise	Depolarizing noise		
		Error rate 0.1	Error rate 0.2	Error rate 0.3
Propose	0.9005	0.8639	0.8664	0.8327
HQNN-Quanv	0.8320	0.7021	0.6502	0.6059
QNN	0.8350	0.7552	0.6944	0.5904

- **Impact of Depolarizing Noise**: The depolarizing channel, as a more comprehensive quantum noise model, significantly affects the existing models. However, the proposed model achieves an accuracy of 0.8327 at an error rate of 0.3, which notably surpasses the performance of HQNN-Quanv (0.6059) and QNN (0.5904).

The findings indicate that the proposed model demonstrates enhanced robustness in the presence of various common noise disturbances, particularly qubit bit-flip and depolarizing noise. This attribute is of paramount significance for the practical applications of quantum computing, where noise is prevalent.

5 | Conclusion

Specifically, we show that training performance is jointly influenced by expressibility, entangling capability, and gradient discreteness, and that high expressibility or strong entanglement alone does not guarantee effective optimization, particularly in the presence of barren plateaus. Furthermore, the parallel architecture is shown to benefit from the global nonlinear transformations induced by the quantum branch, which complement the local receptive fields of classical convolutions and collectively enhance feature discrimination. Finally, the model demonstrates strong robustness under various noise conditions, which is theoretically attributed to the redundancy introduced by parallel quantum–classical feature channels and the use of shallow, hardware-efficient PQCs that mitigate noise effects on NISQ devices. These results indicate that QC-CNN-Parallel offers a principled and practically viable pathway for constructing robust and trainable hybrid quantum learning models.

In summary, the advantages of QC-CNN-Parallel arise not from isolated empirical observations but from the interplay of its parallel convolutional structure, quantum kernel design, and multimetric PQC selection strategy. Nevertheless, the model still presents limitations that warrant future investigation.

5.1 | Future Work

- The current model has not yet been evaluated on large-scale, complex, or multichannel datasets. Future work will focus on further optimizing the QC-CNN-Parallel architecture—both in the quantum and classical convolutional components—to achieve more efficient feature fusion and circuit design, and to assess scalability and robustness in higher-dimensional and multimodal tasks.
- The noise analysis in this work is simulator-based and does not fully capture correlated noise present in real quantum hardware. With continued improvements in qubit counts and hardware fidelity, we plan to implement and evaluate the model on physical quantum devices to further examine its hardware adaptability and real-world performance.

Author Contributions

H.L. conceived the study, designed the methodology, conducted the experiments, performed data analysis, and drafted the original manuscript. X.L. supervised the research, contributed to the conceptual framework and methodology, reviewed and edited the manuscript, and acted as the corresponding author.

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Disclosure

All authors have read and approved the final manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in the MNIST repository at https://yann.lecun.com/exdb/mnist/, the Fashion-MNIST repository at https://github.com/zalandoresearch/fashion-mnist, and the Overhead-MNIST repository at https://github.com/reveondivad/ov-mnist. No additional data were generated during this study.

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Appendix A

(See Table A1).

TABLE A1 | The glossary of abbreviations used in this study.

Abbreviation	Full name	Description
QML	Quantum machine learning	Machine learning models enhanced by quantum computation.
NISQ	Noisy Intermediate-Scale quantum	Quantum computing era characterized by quantum devices with limited qubits and noise.
CNN	Parameterized quantum circuit	Classical deep neural network with convolutional layers for feature extraction.
QNN	Parameterized quantum neural network	Quantum neural network with trainable parameters
HQNN	Hybrid quantum neural network	Hybrid architecture integrating classical neural network layers with quantum circuits.
QC-CNN	Quantum convolutional neural network	Fully quantum CNN where convolution and pooling are implemented by quantum circuits.
QC-CNN	Quantum–classical convolutional neural network	Hybrid quantum–classical model integrating quantum circuits into CNN architecture.
PQC	Parameterized quantum circuit	Quantum circuit with trainable parameters.
HQNN-Parallel	Hybrid quantum neural network with parallel quantum dense layers	Hybrid architecture with parallel quantum dense layers.
HQNN-Quanv	Hybrid quantum neural network with quanvolutional layer	Hybrid model incorporating a quanvolutional layer for quantum feature extraction.